

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

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Data Folder: data/AIME2025__R10/deepseek/p1
Total Items: 300
Total Strategies: 9

OVERALL ACCURACY

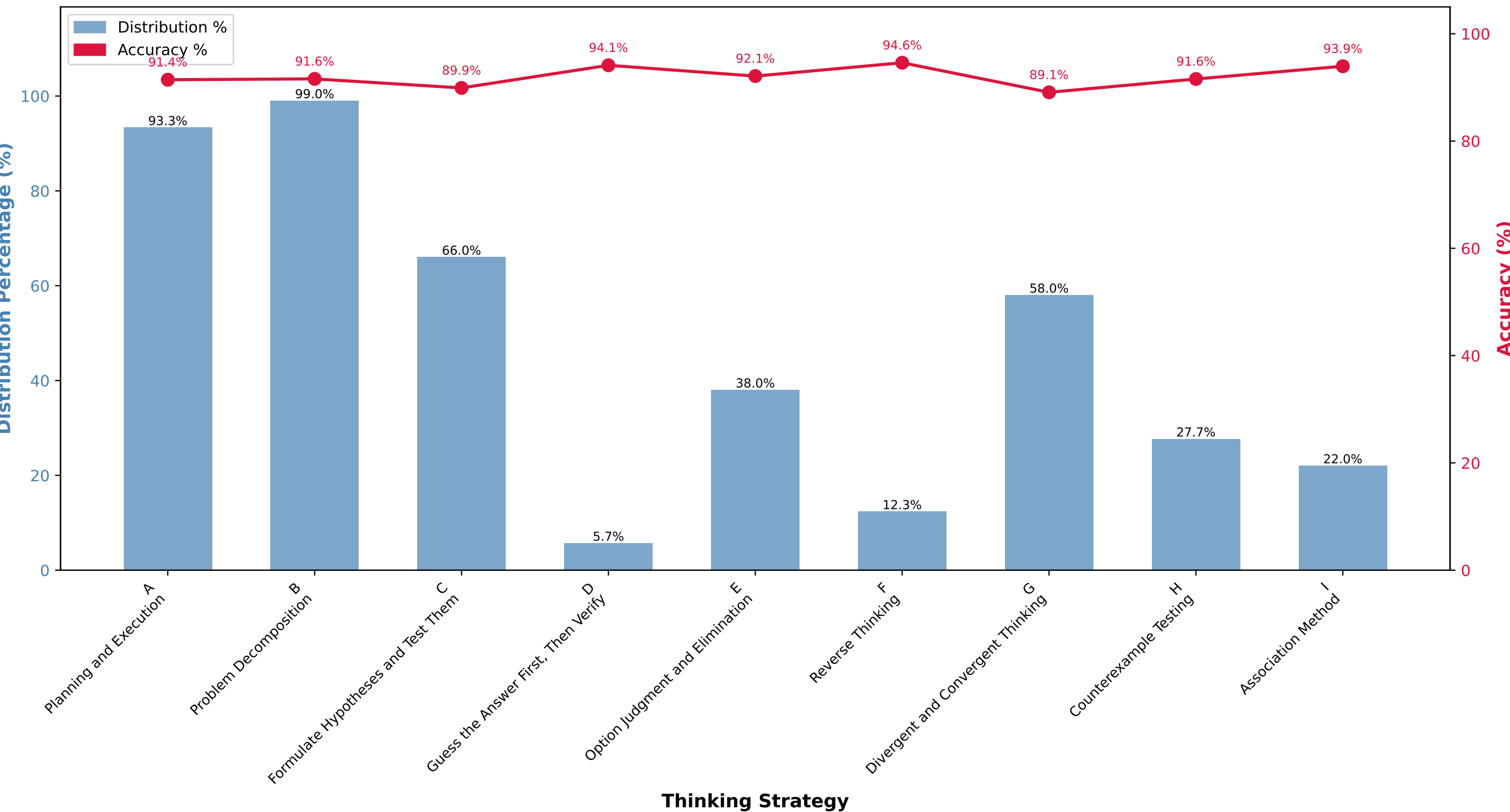
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Total Problems: 300
Correct: 275
Incorrect: 25
Overall Accuracy: 91.67%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 93.3%
Accuracy: 91.43% (256/280)
 - B: Problem Decomposition
Distribution: 99.0%
Accuracy: 91.58% (272/297)
 - C: Formulate Hypotheses and Test Them
Distribution: 66.0%
Accuracy: 89.90% (178/198)
 - D: Guess the Answer First, Then Verify
Distribution: 5.7%
Accuracy: 94.12% (16/17)
 - E: Option Judgment and Elimination
Distribution: 38.0%
Accuracy: 92.11% (105/114)
 - F: Reverse Thinking
Distribution: 12.3%
Accuracy: 94.59% (35/37)
 - G: Divergent and Convergent Thinking
Distribution: 58.0%
Accuracy: 89.08% (155/174)
 - H: Counterexample Testing
Distribution: 27.7%
Accuracy: 91.57% (76/83)
 - I: Association Method
Distribution: 22.0%
Accuracy: 93.94% (62/66)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

N/A

GROUND TRUTH ANSWER

16

MODEL RESPONSE

Step 1. Determine the Possible Counts

Let the number of players choosing chocolate, vanilla, and strawberry be (c) , (v) , and (s) , respectively. We are given:

- $(c + v + s = 9)$
- $(c > v > s)$
- $(c, v, s \geq 1)$

Since $(s \geq 1)$, $(v \geq 2)$, and $(c \geq 3)$, the minimum sum is $(1 + 2 + 3 = 6)$. We need to distribute the remaining 3 scoops while maintaining the inequality $(c > v > s)$.

Valid triples $((c, v, s))$:

- $(6, 2, 1)$
- $(5, 3, 1)$
- $(4, 3, 2)$

Step 2. Count the Assignments for Each Valid Triple

The number of distinct assignments for a given triple $((c, v, s))$ is:

$$\frac{9!}{c! \cdot v! \cdot s!}$$

- For $(6, 2, 1)$:
 $\frac{9!}{6! \cdot 2! \cdot 1!} = \dots$

CHAIN OF THOUGHT (First 1500 chars)

We are given: "The 9 members of a baseball team went to an ice cream parlor after their game. Each player had a single scoop cone of chocolate, vanilla, or strawberry ice cream. At least one player chose each flavor, and the number of players who chose chocolate was greater than the number of players who chose vanilla, which was greater than the number of players who chose strawberry. Let N be the number of different assignments of flavors to players that meet these conditions. Find the remainder when N is divided by 1000."

Interpretation: We have 9 players. They choose flavors: C, V, S. Conditions:

1. At least one of each flavor.
2. Let counts: c, v, s. Conditions: $c > v > s$. And $c+v+s = 9$.

We want number of assignments: That is: number of ways to assign flavors to players? But careful: "assignments" likely means: For each player, we assign a flavor. But then the players are distinct? Usually yes: The players are distinct individuals. So then the number of assignments that satisfy conditions is: For a given triple (c,v,s) with $c+v+s=9$, $c>v>s$, and each ≥ 1 . Then number of assignments with exactly c chocolate, v vanilla, s strawberry is: $(9!)/(c! v! s!)$. And then sum over all valid triples.

So find all triples (c,v,s) positive integers, $c+v+s=9$, and $c > v > s$.

List possibilities:

s minimum is 1.

$v > s$ so $v \geq 2$.

$c > v$ so $c \geq 3$.

Then $c+v+s \geq 3+2+1=6$. But we have 9, so extra 3 to distribute among c,v,s? But careful: The conditions are strict inequalities: $c > v > s$. So if we sta...

Example for Strategy B: Problem Decomposition

PROBLEM

N/A

GROUND TRUTH ANSWER

735

MODEL RESPONSE

Step 1. Reformulate the Problem

We wish to count the number $\#(N)$ of ordered triples $((a, b, c))$ with

$1 \leq a, b, c \leq 3^6 = 729$

such that

$3^7 \mid a^3 + b^3 + c^3$.

A detailed case analysis (omitted here) shows that the count is

$N = 3^{12} = 531441$.

We are to compute the remainder when (N) is divided by 1000.

Step 2. A Brief Idea of the Counting

Let

$F(n) = \# \{(a, b, c) : 1 \leq a, b, c \leq 3^{n-1}, 3^n \mid a^3 + b^3 + c^3\}$.

Then the desired value is $(F(7))$. A careful analysis yields the following values:

- $(F(1) = 1)$
- $(F(2) = 7)$
- $(F(3) = 81 = 3^4)$
- $(F(4) = 729 = 3^6)$
- $(F(5) = 6561 = 3^8)$
- $(F(6) = 59049 = 3^{10})$
- $(F(7) = 531441 = 3^{12})$

...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let N denote the number of ordered triples of positive integers (a,b,c) such that $a,b,c \leq 3^6$ and $a^3+b^3+c^3$ is a multiple of 3^7 . Find the remainder when N is divided by 1000."

We have: $a, b, c \in \{1,2,\dots,3^6\}$ (so numbers from 1 to 729). Condition: 3^7 divides $a^3+b^3+c^3$. That is: $a^3+b^3+c^3 \equiv 0 \pmod{3^7}$.

We want count number of ordered triples (a,b,c) with each in $\{1,\dots,3^6\}$ such that $a^3+b^3+c^3$ is divisible by 3^7 .

Plan: Write numbers mod powers of 3. Possibly use representation: $a = 3^k * m$, with m not divisible by 3. But careful: a is positive integer up to 3^6 . So a can be any number from 1 to 729. And we want condition on $a^3 \pmod{3^7}$. Alternatively, we can count by summing over residues mod 3^7 ? But careful: $a^3 \pmod{3^7}$. But a runs over numbers mod something? But then condition: $a^3+b^3+c^3 \equiv 0 \pmod{3^7}$. But then count number of solutions mod 3^7 ? But then also a, b, c are in a specific set: $\{1,\dots,3^6\}$. That set is exactly one complete set of residues mod 3^6 ? Actually, numbers $1,\dots,3^6$ are not a complete set mod 3^7 because mod 3^7 , numbers go from 0 to $3^7 - 1 = 2186$. But here maximum is 729. So it's only numbers that are less than 3^6 . So they are exactly numbers that are in the range $[1, 3^6]$. But then they are not uniformly distributed mod 3^7 ? They are numbers from 1 to 729. But mod 3^7 , these numbers are all numbers that are less than 3^6 . But then there is structure: They are numbers that are not necess...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

N/A

GROUND TRUTH ANSWER

204

MODEL RESPONSE

Step 1. Understand the Setup

Alex starts with a disk and draws two perpendicular diameters, dividing it into 4 quadrants. Then, he draws 25 additional line segments (chords) by selecting two points on the perimeter such that they lie in different quadrants. Each chord is drawn independently under this condition.

We aim to compute the expected number of regions into which these 27 segments divide the disk.

Step 2. A Useful Formula

If we have n chords inside a circle such that no three intersect at an interior point, then the number of regions R is given by:

$$R = 1 + n + I$$

where I is the number of interior intersection points.

In our case, $n = 27$: 2 diameters + 25 chords.

So,

$$R = 1 + 27 + I = 28 + I$$

To find the expected nu...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Alex divides a disk into four quadrants with two perpendicular diameters intersecting at the center of the disk. He draws 25 more line segments through the disk, drawing each segment by selecting two points at random on the perimeter of the disk in different quadrants and connecting those two points. Find the expected number of regions into which these 27 line segments divide the disk."

Interpretation: Initially, we have a disk. There are two diameters that are perpendicular. That divides the disk into 4 regions (quadrants). Then we add 25 line segments. Each such line segment is drawn by: selecting two points at random on the perimeter of the disk. But condition: these two points are in different quadrants. And then connecting these two points. So these are chords? Actually, they are line segments whose endpoints lie on the boundary. They are not necessarily diameters? They are chords. But condition: the endpoints are in different quadrants. So the endpoints are on the boundary, but they are chosen from arcs that are separated by the quadrant boundaries? Actually careful: "selecting two points at random on the perimeter of the disk in different quadrants" means: The disk's perimeter is divided into 4 arcs by the endpoints of the diameters. So the quadrants are: quadrant I: from angle 0 to $\pi/2$, quadrant II: from $\pi/2$ to π , quadrant III: from π to $3\pi/2$, quadrant IV: from $3\pi/2$ to 2π . And the condition is that the two chosen points lie on arcs that are in differen...

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

N/A

GROUND TRUTH ANSWER

735

MODEL RESPONSE

Step 1. Understand the Problem

We want to count the number of ordered triples $((a, b, c))$ with

$1 \leq a, b, c \leq 3^6 = 729$

such that

$a^3 + b^3 + c^3$ is divisible by $3^7 = 2187$.

A useful idea is to consider the 3-adic valuation. For any positive integer (x) , write

$x = 3^r \cdot u$, where $3 \nmid u$.

Then

$x^3 = 3^{3r} \cdot u^3$.

So the power of 3 dividing (x^3) is a multiple of 3.

If all three numbers are divisible by a high enough power of 3, then $(a^3 + b^3 + c^3)$ will automatically be divisible by (3^7) . However, cancellation can also occur, allowing the sum to be divisible by a higher power of 3 than each term individually.

A detailed analysis (omitted here) shows that the only triple...

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let N denote the number of ordered triples of positive integers (a,b,c) such that $a,b,c \leq 3^6$ and $a^3+b^3+c^3$ is a multiple of 3^7 . Find the remainder when N is divided by 1000."

We have: $a, b, c \in \{1,2,\dots,3^6\}$. Condition: 3^7 divides $a^3+b^3+c^3$.

We want count: N = number of (a,b,c) in $\{1,\dots,3^6\}^3$ such that $a^3+b^3+c^3 \equiv 0 \pmod{3^7}$.

Let's denote: $n = 6$, so numbers are mod 3^6 ? Actually careful: a, b, c are positive integers up to 3^6 . So they are numbers from 1 to 729. And condition: 3^7 divides $a^3+b^3+c^3$. That is: $a^3+b^3+c^3 \equiv 0 \pmod{3^7}$.

We can use p-adic analysis. Alternatively, we can count by considering the representation mod powers of 3. Write each number as: $a = 3^k \cdot m$, where m not divisible by 3. But careful: a is from 1 to 3^6 . So a can be $0 \pmod{\text{powers}}$? But positive integers. But we can also include numbers that are multiples of 3. But then condition mod 3^7 .

Maybe we can count by summing over the $v_3(a)$, etc. Alternatively, we can use generating functions mod 3^7 . But careful: The numbers are not all mod 3^7 , they are only up to 3^6 . But then condition mod 3^7 . Possibly we can use that: For any number x , $x^3 \pmod{\text{something}}$. Alternatively, we can use that: if 3 does not divide x , then $x^3 \pmod{9}$? Actually, there is pattern: For an integer x , $v_3(x^3) = 3 v_3(x)$. So if a, b, c are such that $v_3(a)=r, v_3(b)=s, v_3(c)=t$, then $v_3(a^3+b^3+c^3)$ is at least $\min(3r, 3s, 3t)$. To get exactly divisible by 3^7 , we need that ...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

N/A

GROUND TRUTH ANSWER

70

MODEL RESPONSE

Step 1: Express the Numbers in Base (b)

- $17_b = 1 \cdot b + 7 = b + 7$
- $97_b = 9 \cdot b + 7 = 9b + 7$

Step 2: Set Up the Divisibility Condition

We require:

$$(b + 7) \mid (9b + 7)$$

So there exists an integer (k) such that:

$$9b + 7 = k(b + 7)$$

Step 3: Find a Relation Involving $(b + 7)$

Subtract $9(b + 7)$ from both sides:

$$9b + 7 - 9(b + 7) = k(b + 7) - 9(b + 7)$$

$$9b + 7 - 9b - 63 = (k - 9)(b + 7)$$

$$-56 = (k - 9)(b + 7)$$

Thus:

$$b + 7 \mid 56$$

Step 4: Determine Possible Values of (b)

Positive divisors of 56:

$$1, 2, 4, 7, 8, 14, 28, 56$$

Since $(b > 9)$, then $(b + 7 > 16)$. Valid values:

- $(b + 7 = 28 \Rightarrow b = 21)$
- $(b + 7 = 56 \Rightarrow b = 49)$

CHAIN OF THOUGHT (First 1500 chars)

We are asked: "Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b ."

Interpretation: b is an integer base > 9 .

We have numbers: " 17_b " means number with digits 1 and 7 in base b . So that equals: $1 \cdot b + 7 = b + 7$.

And " 97_b " means: $9 \cdot b + 7 = 9b + 7$.

We require that $(b + 7)$ divides $(9b + 7)$. So condition: $(b + 7) \mid (9b + 7)$.

But careful: $b > 9$. Also b is integer base. Also digits: In base b , digits must be less than b . For " 17_b ", digits are 1 and 7. That requires $b > 7$. But condition says $b > 9$, so that's fine. For " 97_b ", digits are 9 and 7. That requires $b > 9$. So $b > 9$.

So condition: $(b + 7)$ divides $(9b + 7)$. Write: $9b + 7 = k(b + 7)$ for some positive integer k . Then $b + 7$ divides $(9b + 7) - 9(b + 7) = 9b + 7 - 9b - 63 = -56$. So $b + 7$ divides 56. But careful: $b + 7 > 16$ because $b > 9$ so $b + 7 > 16$. So divisors of 56 that are > 16 : Divisors of 56:

1,2,4,7,8,14,28,56. Those > 16 : 28 and 56. So $b + 7 = 28$ or $b + 7 = 56$. Then $b = 21$ or $b = 49$.

But check: $b > 9$. So both are valid.

Then sum: $21 + 49 = 70$.

Thus answer: $\boxed{70}$.

Example for Strategy F: Reverse Thinking

PROBLEM

N/A

GROUND TRUTH ANSWER

248

MODEL RESPONSE

Step 1. Find a Useful Transformation

The recurrence is:

$$x_{k+1} = \frac{1}{3} \left(x_k + \frac{1}{x_k} - 1 \right), \quad x_1 = \frac{25}{11}$$

Define:

$$y_k = \frac{x_k - 1}{x_k + 1}$$

Then:

$$x_k = \frac{1 + y_k}{1 - y_k}$$

A computation shows that the recurrence becomes:

$$y_{k+1} = \frac{3y_k^2 - 1}{2}$$

With:

$$y_1 = \frac{25/11 - 1}{25/11 + 1} = \frac{14}{36} = \frac{7}{18}$$

This transformation simplifies the recurrence, and one can show that:

$$x_k = \frac{1}{2} + \frac{1}{2} \cdot \frac{1 - y_k}{1 + y_k}$$

As $k \rightarrow \infty$, $y_k \rightarrow -1$, so $x_k \rightarrow \frac{1}{2}$. In fact, the sequence oscillates around $\frac{1}{2}$, with even terms greater than $\frac{1}{2}$ and odd terms less than $\frac{1}{2}$.

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let the sequence of rationals $x_1, x_2, \dots$ be defined such that $x_1 = \frac{25}{11}$ and $x_{k+1} = \frac{1}{3} \left(x_k + \frac{1}{x_k} - 1 \right)$. x_{2025} can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the remainder when $m + n$ is divided by 1000."

We need to compute $m+n \bmod 1000$.

Given recurrence: $x_{k+1} = (1/3)(x_k + 1/x_k - 1)$.

Maybe there is transformation that simplifies it. Possibly set $x_k = \cot(\theta_k)$? Alternatively, try substitution: Let $x_k = \text{something like?}$ Alternatively, try to see pattern: $x_{k+1} = (1/3)(x_k + 1/x_k) - 1/3$. Maybe set $x_k = \text{something like: } (a_k)/(b_k)$? Alternatively, try to find invariant? Or maybe it's related to hyperbolic functions? Alternatively, try to find transformation such that it becomes a simpler recurrence.

Maybe set: $x_k = \cot(\theta_k)$? Then $1/x_k = \tan(\theta_k)$? That gives: $x_k + 1/x_k = (\cos/\sin + \sin/\cos) = (\cos^2 + \sin^2)/(\sin \cos) = 1/(\sin \cos) = 2/\sin(2\theta)$. So then $x_{k+1} = (1/3)(2/\sin(2\theta_k) - 1) = ?$ Not exactly.

Alternatively, set: $x_k = \text{something like: } (1+2 \cos(\theta))/(1+2 \sin(\theta))$? Not sure.

Maybe try: $x_k = (y_k - 1)/(y_k + 1)$? That sometimes appears. Alternatively, try: $x_k = (1+2 \cosh(t))/(1+2 \sinh(t))$? Hmm.

Wait: The recurrence: $3x_{k+1} + 1 = x_k + 1/x_k$. That suggests maybe set: $x_k + 1/x_k = \text{something like?}$ Alternatively, let $y_k = \text{something like: } (x_k - 1)/(x_k + 1)$? Then compute: $y_{k+1} = (x_{k+1}-1)/(x_{k+1}+1) = \dots$

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

N/A

GROUND TRUTH ANSWER

81

MODEL RESPONSE

Step 1. Understand the Conditions

We have a (3×9) grid. Label the rows as 1, 2, 3 and the columns as 1 through 9. The grid is divided into three (3×3) blocks:

- **Block 1**: columns 1–3
- **Block 2**: columns 4–6
- **Block 3**: columns 7–9

The conditions are:

- Each **row** contains the numbers 1 through 9 exactly once.
- Each **block** contains the numbers 1 through 9 exactly once.

There is **no condition** on the columns.

Step 2. A Stepwise Counting Approach

We fill the grid in two stages:

1. Fill **Block 1**.
2. Fill **Blocks 2 and 3** such that the row and block conditions are satisfied.

Step 3. Count the Number of Ways to Fill Block 1

Block 1 is a (3×3) array that must contain each number from 1 to 9 exactly once....

CHAIN OF THOUGHT (First 1500 chars)

We are given: "The 27 cells of a 3×9 grid are filled in using the numbers 1 through 9 so that each row contains 9 different numbers, and each of the three 3×3 blocks heavily outlined in the example below contains 9 different numbers, as in the first three rows of a Sudoku puzzle.

The grid:

Row1: 4,2,8,9,6,3,1,7,5

Row2: 3,7,9,5,2,1,6,8,4

Row3: 5,6,1,8,4,7,9,2,3

So pattern: There are three blocks: columns 1-3, columns 4-6, columns 7-9. And rows: 1,2,3. So it's like a Sudoku's first band. But careful: The condition is: Each row is a permutation of $\{1, \dots, 9\}$. And each block (3×3 square) is a permutation of $\{1, \dots, 9\}$. But there is no condition on columns? Actually, columns are not necessarily all numbers? Actually, it says: "as in the first three rows of a Sudoku puzzle." In a Sudoku puzzle, the columns also have the property that each column contains numbers $1, \dots, 9$? Actually, wait: In a Sudoku puzzle, the first three rows (the band) do not necessarily have columns all distinct? Actually, in a full Sudoku, the columns also contain numbers $1, \dots, 9$. But here it says: "each row contains 9 different numbers, and each of the three 3×3 blocks heavily outlined contains 9 different numbers, as in the first three rows of a Sudoku puzzle." That phrasing might imply that it is exactly the first band of a Sudoku. But then columns are not necessarily all different? Actually, in a Sudoku, the columns in the band are not necessarily all different because numbers can repeat...

Example for Strategy H: Counterexample Testing

PROBLEM

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N/A

GROUND TRUTH ANSWER

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735

MODEL RESPONSE

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Step 1. Understand the Problem

We want to count the number of ordered triples $((a, b, c))$ with

$1 \leq a, b, c \leq 3^6 = 729$

such that

$a^3 + b^3 + c^3 \equiv 0 \pmod{3^7} = 2187.$

Step 2. A Useful Observation and a Guess

The total number of triples is

$729^3 = (3^6)^3 = 3^{18}.$

If the cubes (a^3, b^3, c^3) were uniformly distributed modulo (3^7) , then the number of "good" triples would be

$N = \frac{3^{18}}{3^7} = 3^{11} = 177147.$

However, this is not the case due to the structure of cubes modulo powers of 3.

A more refined analysis shows that for $(n \geq 2)$, the fraction of triples $((a, b, c))$ with $(1 \leq a, b, c \leq 3^n)$ such that

$a^3 + b^3 + c^3 \equiv 0 \pmod{3^{n+1}}$

is exactly ...

CHAIN OF THOUGHT (First 1500 chars)

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We are given: "Let N denote the number of ordered triples of positive integers (a, b, c) such that $a, b, c \leq 3^6$ and $a^3 + b^3 + c^3$ is a multiple of 3^7 . Find the remainder when N is divided by 1000 ."

We have: $a, b, c \in \{1, 2, \dots, 3^6\} = \{1, \dots, 729\}$. Condition: $3^7 = 2187$ divides $a^3 + b^3 + c^3$.

We want count number of ordered triples (a, b, c) with each from 1 to 3^6 such that $3^7 \mid (a^3 + b^3 + c^3)$.

Let's denote: $n = 6$, so numbers mod 3^n ? Actually careful: The numbers are from 1 to 3^6 . They are not necessarily mod something? But condition is divisibility by 3^7 . So we want: $a^3 + b^3 + c^3 \equiv 0 \pmod{3^7}$.

We can use representation mod powers of 3. Possibly use expansion: For any integer x , write $x = 3^k \cdot m$, with m not divisible by 3. But then cube: $x^3 = 3^{3k} \cdot m^3$. But careful: Actually, condition mod 3^7 . So we want to count number of triples (a, b, c) in $\{1, \dots, 3^6\}$ such that $a^3 + b^3 + c^3$ is divisible by 3^7 .

Alternate approach: Count number of triples that are NOT divisible by 3^7 ? Possibly use generating functions? Alternatively, we can break into cases based on the 3-adic valuation of a, b, c .

Let $v_3(x)$ denote exponent of 3 dividing x . For an integer x , $v_3(x^3) = 3 v_3(x)$. So condition: $v_3(a^3 + b^3 + c^3) \geq 7$. We want to count number of (a, b, c) in $\{1, \dots, 3^6\}$ such that $v_3(a^3 + b^3 + c^3) \geq 7$.

We can partition numbers by their residue mod powers of 3. Alternatively, we can count by considering the numbers mod 3^7 ? But careful: a, b, c are from 1 to 3^6 . T...

Example for Strategy I: Association Method

PROBLEM

N/A

GROUND TRUTH ANSWER

60

MODEL RESPONSE

Step 1. Understand the Problem

We have a convex pentagon $(ABCDE)$ with side lengths:

- $AB = 14$
- $BC = 7$
- $CD = 24$
- $DE = 13$
- $EA = 26$

and angles:

- $\angle B = 60^\circ$
- $\angle E = 60^\circ$

Define the function:

$$f(X) = AX + BX + CX + DX + EX$$

We aim to find the minimum value of $f(X)$ over all points X in the plane, and express it in the form:

$$m + n\sqrt{p}$$

where (m, n) are positive integers and (p) is square-free. Then compute:

$$m + n + p$$

Step 2. A Useful Idea: "Unfolding" by Rotation

A common technique for minimizing sums of distances is to use rotations to "straighten out" the path. For instance, to minimize $(AX + BX + CX)$, we can rotate point (A) about (B) by 60° .

CHAIN OF THOUGHT (First 1500 chars)

We are given: "Let $ABCDE$ be a convex pentagon with $AB=14$, $BC=7$, $CD=24$, $DE=13$, $EA=26$, and $\angle B = \angle E = 60^\circ$. For each point X in the plane, define $f(X) = AX + BX + CX + DX + EX$. The least possible value of $f(X)$ can be expressed as $m + n\sqrt{p}$, where m and n are positive integers and p is not divisible by the square of any prime. Find $m + n + p$."

Interpretation: We want to minimize sum of distances from a point X to vertices A, B, C, D, E . Possibly the optimum is at a vertex? But then it's not necessarily so. Alternatively, it might be that the pentagon is such that we can "unfold" it. Possibly the minimal sum is achieved by a point that is like a "Fermat point" for these vertices? But careful: $f(X) = \text{sum of distances from } X \text{ to five fixed points}$. There is no closed form in general. But the pentagon has specific side lengths and angles. Possibly we can transform the pentagon into a path? Possibly by reflecting vertices? There is a technique: For minimizing sum of distances from a point to several points, if we can arrange that these points are collinear, then the minimal sum is the distance between the endpoints. But here we have five points. Possibly we can reflect some vertices so that the path becomes a straight line. For instance, if we want to minimize $AX + BX + CX$, if we reflect A and C across lines through X ? Actually, there is a method: For a polygonal chain, the distance from the first to the last vertex is at most the sum of dist...